

$$\langle f|g \rangle = \int_1^5 f(x) g(x) dx$$

$$\mathcal{H}_1 = [1, 5] \quad b = \begin{pmatrix} 1|x_1 & x_2 \\ \vec{n}_1 & \vec{n}_2 & \vec{n}_3 \end{pmatrix}$$

$$a) \quad \vec{n}_1 = \text{normalizovaná} \quad \vec{n}_1 = \frac{\vec{n}_1}{\|\vec{n}_1\|} = \frac{1}{2}$$

$$\|\vec{n}_1\| = \sqrt{\int_1^5 1 dx} = 2$$

$$\vec{n}_2 = \langle w_1 | w_2 \rangle w_1 = 6 \cdot \frac{1}{2} = 3$$

$$\langle \vec{n}_1, \vec{n}_2 \rangle = \int_1^5 \frac{x}{2} dx = 6$$

$$\vec{n}_2 \leftarrow x \quad \vec{n}_2 - \vec{n}_2 = x - 3$$

$$\|\vec{n}_2 - \vec{n}_2\| = \sqrt{\int_1^5 (x-3)^2 dx} = \sqrt{\frac{16}{3}}$$

$$\vec{n}_2 = \text{normalizovaná kolmice} \quad \frac{\vec{n}_2 - \vec{n}_2}{\|\vec{n}_2 - \vec{n}_2\|} = \frac{\sqrt{3}(x-3)}{\sqrt{16}}$$

$$\vec{n}_3 = \langle \vec{n}_1, \vec{n}_3 \rangle \vec{n}_1 + \langle \vec{n}_2, \vec{n}_3 \rangle \vec{n}_2 = \frac{31}{3} + \frac{24(x-3)}{\sqrt{16}}$$

$$\langle \vec{n}_1, \vec{n}_3 \rangle = \int_1^5 \frac{x^2}{2} dx = \frac{62}{3}$$

$$\langle \vec{n}_1, \vec{n}_3 \rangle \cdot \vec{n}_1 = \frac{62}{3} \cdot \frac{1}{2} = \frac{31}{3}$$

$$\langle \vec{n}_2, \vec{n}_3 \rangle = \int_1^5 \frac{\sqrt{3}(x-3)}{\sqrt{16}} \cdot x^2 dx = 8\sqrt{3}$$

$$\langle \vec{n}_2, \vec{n}_3 \rangle \vec{n}_2 = \frac{8\sqrt{3} \cdot \sqrt{3}(x-3)}{\sqrt{16}} = \frac{24(x-3)}{\sqrt{16}}$$

$$\|\vec{n}_3 - \vec{n}_3\| = \sqrt{\int_1^5 \left(x^2 - \frac{24(x-3)}{\sqrt{16}} - \frac{31}{3}\right)^2 dx} = \frac{16}{8\sqrt{3}}$$

$$b) \quad b = \left(\frac{1}{2}, \frac{\sqrt{3}(x-3)}{4}, \frac{3\sqrt{5}(x^2 - 6(x-3) - \frac{31}{3})}{16} \right) \text{ podprostor } V \quad \sin x \approx ax^2 + bx + c \quad a, b, c \in \mathbb{R} \quad a, b, c = ?$$

Podle věty 8.54 je b báze pro sin x na podprostoru V vyjádřena takto

$$\begin{pmatrix} \frac{1}{2} \sin x \\ \frac{\sqrt{3}(x-3)}{4} \sin x \\ \frac{3\sqrt{5}(x^2 - 6(x-3) - \frac{31}{3})}{16} \sin x \end{pmatrix} = \begin{pmatrix} \int_1^5 \frac{\sin x}{2} \\ \int_1^5 \frac{\sqrt{3} \sin x (x-3)}{4} \\ \int_1^5 \frac{3\sqrt{5} \sin x (x^2 - 6(x-3) - \frac{31}{3})}{16} \end{pmatrix} \approx \begin{pmatrix} \frac{\cos(1) - \cos(5)}{2} \\ \frac{\sqrt{3}(\sin(1) - \sin(5) + 2\cos(1) + 2\cos(5))}{4} \\ \frac{\sqrt{5}(6(\sin(1) + \sin(5)) + \cos(1) - \cos(5))}{8} \end{pmatrix} \approx \begin{pmatrix} 0,12832 \\ -1,4432 \\ -0,12524 \end{pmatrix} \approx \begin{pmatrix} 0,128 \\ -1,443 \\ -0,125 \end{pmatrix}$$

$$\sin x \approx \frac{\cos(1) - \cos(5)}{4} - \frac{3(x-3)(\sin(1) - \sin(5) + 2\cos(1) + 2\cos(5))}{16} + \frac{15(6(\sin(1) + \sin(5)) + \cos(1) - \cos(5))(x^2 - 6(x-3) - \frac{31}{3})}{128}$$

$$\sin x \approx \frac{\cos(1) - \cos(5)}{4} - \frac{3x(\sin(1) - \sin(5) + 2\cos(1) + 2\cos(5))}{16} + \frac{9(\sin(1) - \sin(5) + 2\cos(1) + 2\cos(5))}{16}$$

$$+ \frac{15x^2(6(\sin(1) + \sin(5)) + \cos(1) - \cos(5))}{128} - \frac{75x(6(\sin(1) + \sin(5)) + \cos(1) - \cos(5))}{128} + \frac{270(6(\sin(1) + \sin(5)) + \cos(1) - \cos(5))}{128}$$

$$- \frac{465(6(\sin(1) + \sin(5)) + \cos(1) - \cos(5))}{384} \approx \frac{15x^2(6(\sin(1) + \sin(5)) + \cos(1) - \cos(5))}{128} - \frac{3x(\sin(1) - \sin(5) + 2\cos(1) + 2\cos(5))}{16} - \frac{75x(6(\sin(1) + \sin(5)) + \cos(1) - \cos(5))}{128}$$

$$+ \frac{\cos(1) - \cos(5)}{4} + \frac{9(\sin(1) - \sin(5) + 2\cos(1) + 2\cos(5))}{16} + \frac{270(6(\sin(1) + \sin(5)) + \cos(1) - \cos(5))}{128} - \frac{465(6(\sin(1) + \sin(5)) + \cos(1) - \cos(5))}{384} \approx *$$

$$+ \frac{3155 \cdot 0,12524}{16} \approx -0,052508466x^2 - 0,331523769x + 1,601378791 \approx -0,053x^2 - 0,332x + 1,601$$

$$c \approx \frac{\cos(1) - \cos(5)}{4} + \frac{9 \cdot (\sin(1) - \sin(5) + 2\cos(1) + 2\cos(5))}{16} + \frac{270 \cdot (6(\sin(1) + \sin(5)) + \cos(1) - \cos(5))}{128} - \frac{465 \cdot (6(\sin(1) + \sin(5)) + \cos(1) - \cos(5))}{384}$$

$$\approx \frac{0,12832}{2} + \frac{3\sqrt{3} \cdot 1,4932}{4} - \frac{5455 \cdot 0,12524}{16} + \frac{3155 \cdot 0,12524}{16} \approx 1,601378791 \approx \underline{\underline{1,601}}$$

$$b \approx \frac{-3(\sin(1) - \sin(5) + 2\cos(1) + 2\cos(5))}{16} - \frac{75 \cdot (6(\sin(1) + \sin(5)) + \cos(1) - \cos(5))}{128}$$

$$\approx \frac{-\sqrt{3} \cdot 1,4932}{4} + \frac{1855 \cdot 0,12524}{16} \approx -0,331523769 \approx \underline{\underline{-0,332}}$$

$$a \approx \frac{15(6(\sin(1) + \sin(5)) + \cos(1) - \cos(5))}{128} \approx -\frac{355 \cdot 0,12524}{16} \approx -0,052508466 \approx \underline{\underline{-0,053}}$$

$$C) ||\sin x - (ax^2 + bx + c)|| = \sqrt{\int_1^5 \left(\sin x - \frac{15x^2(6(\sin(1) + \sin(5)) + \cos(1) - \cos(5))}{128} + \frac{3x(\sin(1) - \sin(5) - 2\cos(1) + 2\cos(5))}{16} + \frac{75x \cdot (6(\sin(1) + \sin(5)) + \cos(1) - \cos(5))}{128} \right.}$$

$$\left. - \frac{\cos(1) - \cos(5)}{4} - \frac{9 \cdot (\sin(1) - \sin(5) + 2\cos(1) + 2\cos(5))}{16} - \frac{270 \cdot (6(\sin(1) + \sin(5)) + \cos(1) - \cos(5))}{128} + \frac{465 \cdot (6(\sin(1) + \sin(5)) + \cos(1) - \cos(5))}{384} \right)^2}$$

$$\approx \sqrt{\int_1^5 \left(\sin x + \frac{365 \cdot 0,12524x^2}{16} + \frac{\sqrt{3} \cdot 1,4932x}{4} - \frac{1855 \cdot 0,12524x}{16} - \frac{0,12832}{2} - \frac{3\sqrt{3} \cdot 1,4932}{4} + \frac{5455 \cdot 0,12524}{16} - \frac{3155 \cdot 0,12524}{16} \right)}$$

$$\approx \sqrt{0,101627} \approx 0,318789899 \approx \underline{\underline{0,319}}$$